

SARTHAK BICHHAWA

Senior Data & Applied Scientist · Machine Learning, Analytics & Applied AI

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SUMMARY

Data scientist with **9+ years** shipping ML, analytics, and AI systems that executives actually use — at Microsoft Azure and Walmart. Takes high-stakes data problems end-to-end: a **\$5B+/year revenue pipeline**, reliability programs for the cloud's largest storage customers, and **production LLM agents** that turned days of reporting into minutes.

HIGHLIGHTS

- Recovered **~\$45M/month** of partner revenue attribution after a critical pipeline failure — restoring the **\$5B+/year** revenue picture leadership relies on.
- Eliminated **99%** of performance-throttling incidents for Azure Storage's most critical customers.
- Built **LLM agents and MCP servers** used daily by leadership and engineers; org-recognized **AI Champion**.

EXPERIENCE

Microsoft — Azure Storage | Senior Data & Applied Scientist

Nov 2020 – Present

Own revenue-critical data platforms, customer-health analytics, and applied-AI systems for one of the world's largest cloud storage services.

- Recovered **~\$45M/month** of partner-attributed revenue after an upstream data loss silently broke attribution for **\$5B+/year** of Azure Storage revenue — traced the failure through a 20+ component legacy pipeline, sourced an alternate telemetry feed, and drove six months of backfills through quarter close (7 incidents, including a **severity-2 I led across 4 teams**).
- Redesigned the attribution engine end-to-end — **multi-partner revenue credit** replacing last-writer-wins (**~80–90% more** partner accounts tracked), validation gates with automated backfills, and independent reconciliation against Finance actuals so the business can trust the numbers.
- Built an **LLM agent** that answers customer-health questions in plain English by querying live telemetry — cutting executive-review prep from **days to under 2 minutes**.
- Shipped an autonomous **multi-agent incident investigator** (Azure OpenAI) that cuts on-call diagnosis from hours to minutes, plus **four reusable MCP servers** — including one that lets **Copilot query any live operations dashboard**.
- Eliminated **99%** of throttling incidents for the platform's most critical customers (92% across all paying customers) by building fleet-wide telemetry pipelines and an early-warning dashboard with automated triage.
- Built a self-service **data-quality platform** — automated **freshness, volume, and schema anomaly checks**, quality gates, and **hourly profiling of 400+ production datasets** — whose first named catch was an **88% silent row loss** headed for financial reports.
- Grew it into an internal **data discovery & governance portal** and designed its core services — including an **LLM-powered lineage engine that reads pipeline code** to map production data flows — launched org-wide so any engineer can answer “can I rely on this data right now?”
- Automated the monthly business review from a multi-day manual process into a governed **~5-minute pipeline** (**>95% less effort**, reconciles automatically against Finance actuals) that leadership now consumes directly.
- Designed a **simulation-based stress-testing program** running thousands of controlled experiments per release, with multi-armed bandits prioritizing tests — prevented **10+ major customer incidents** and cut test-infrastructure cost **40%**.
- Drove the executive health-review program for the largest storage customers across **8+ engineering teams** — **50% fewer** long-aging issues, **16% faster** mitigation, helping retain **10+ accounts** worth millions per month each.
- Recognized as the org's **AI Champion** — taught a hands-on internal workshop on agent-based automation and mentored engineers across the team.

Walmart Labs | Senior Data Scientist

Feb 2019 – Nov 2020

- Delivered **~7% category revenue lift** across 50 stores with an end-to-end shelf-layout optimization system, cutting planning cycles from weeks to hours.
- Led a team of **4 data scientists + 2 interns** to productionize the workflow; earned a Spot Award.

Integral Ad Science | Data Scientist Intern

Jun 2018 – Aug 2018

- Built a PySpark text-classification pipeline processing **~1M URLs/day** across 846 categories at 88% accuracy.

Mu Sigma | Data Scientist

Oct 2015 – Jul 2017

- Modeled **customer lifetime value** with probabilistic/regression models for Fortune-500 clients.

SKILLS

Machine Learning & AI: machine learning, generative AI, LLM agents, retrieval-augmented generation (RAG), Model Context Protocol (MCP), Azure OpenAI, embeddings, anomaly detection, A/B testing, causal inference, multi-armed bandits, statistical modeling

Engineering & Data: Python (pandas, scikit-learn, PyTorch, XGBoost), SQL, Spark/PySpark, FastAPI, Azure Data Explorer, Azure Data Factory, Power BI, Azure DevOps, Git, data pipelines & governance

EDUCATION

University of Illinois at Chicago | MS, Business Analytics

2017 – 2018

SGSITS, Indore | BE, Computer Engineering

2011 – 2015